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## CY 643: Advanced Analytical Techniques

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| Course Code  | CY 643                         |
| Course Name  | Advanced Analytical Techniques |
| L-T-P-C      | 3-0-0-3                        |
| Intended for | B.Tech/MSc/PhD                 |
| Distribution | Compulsory                     |
| Approval     | 1st Adhoc Senate (Apr 2011)    |

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### Course Contents

- **Introduction**

Role of Analytical Chemistry in Science, Classification of quantitative methods in the science, steps in typical quantitative analysis. [2 Lectures]

- **Surface Analysis by Microscopy**

Scanning Electron Microscopy (SEM), Atomic Force Microscopy (AFM)-Principle, Instrumentation, Analysis, and Applications. [10 Lectures]

- **Atomic Spectroscopy for Analysis**

Atomic Absorption and Atomic Emission Spectroscopy - Principle, Instrumentation, Analysis, and Applications and Advances. [8 Lectures]

- **Purification and Isolation techniques**

An introduction to chromatographic techniques, Gas Liquid chromatography and Liquid chromatography (particularly High Performance Liquid Chromatography) - Principles, Instrumentation, Applications and Advances. [8 Lectures]

- **Miscellaneous Methods of Analysis**

Thermal Methods- Differential Thermal Analysis, Differential Scanning Calorimetry. Particle Size Determination Dynamic Light Scattering. [14 Lectures]

### Text and Reference Books

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1. Skoog, Holler, Crouch, **Instrumental Analysis**.
2. Skoog, West, Holler, **Fundamentals of Analytical Chemistry**